

Software Requirements Specification

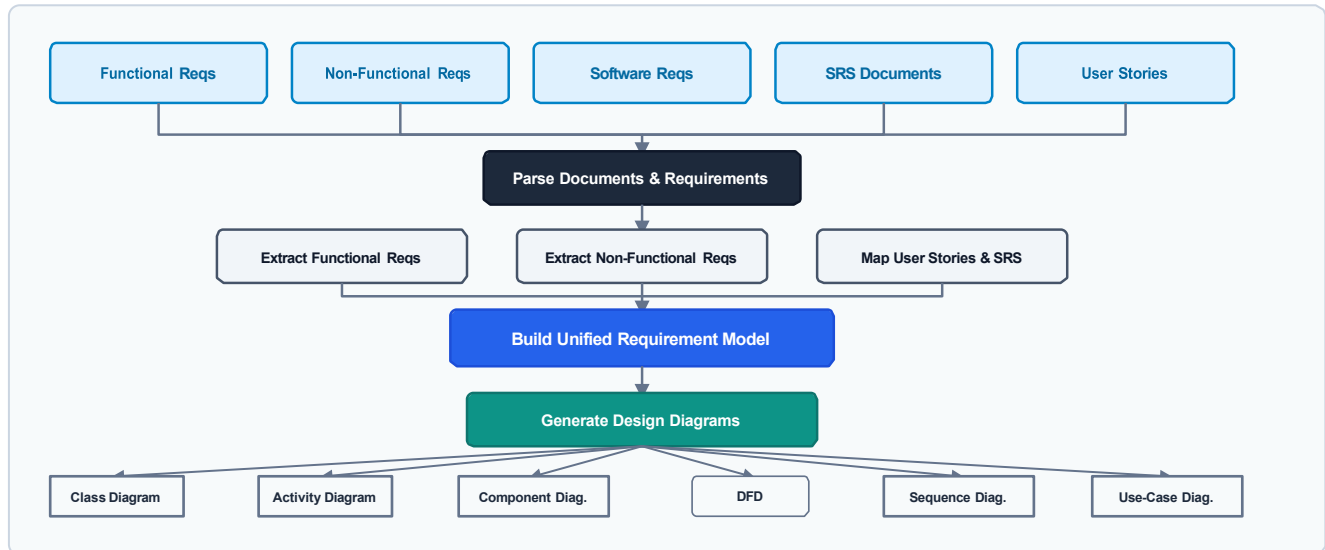
Software Design Diagram Generator (SDDG)

Course: IT31 4 Software Engineering

Group: Group 27

Coverage: FR1 –FR30 | NFR1 –NFR1 0 |
DR1 –DR2

SYSTEM ARCHITECTURE & PROCESSING PIPELINE



1. INTRODUCTION & SYSTEM SCOPE

1.1 Purpose: The Software Design Diagram Generator (SDDG) is a web application that turns natural-language software requirements into six types of software design diagrams, allowing users to review, edit, version, and trace diagrams back to source requirements.

1.2 System Scope Boundaries

In Scope: English text/PDF/DOCX documents, user stories, FR/NFR lists; automated generation of Use Case, Class, Activity, Sequence, DFD, and Component diagrams (exportable as PNG, SVG, PDF); requirement extraction, validation, editing, versioning, traceability, and collaborative admin controls on desktop Chrome, Firefox, and Edge.

Out of Scope: OCR for scanned image-only PDFs; separate flowchart diagrams (Activity diagrams serve process-flow needs); real-time simultaneous co-editing; mobile/tablet layouts; non-English inputs; third-party login providers other than Google; importing pre-existing external diagrams.

1.3 DEFINITIONS & SYSTEM TERMINOLOGY

Term	Definition & Operational Rules
Project	A named workspace owned by one or more users that holds input sources, requirement statements, extracted items, diagrams, and their full history.
Document / Page	Uploaded PDF or DOCX file (up to 1.5 per project). Text/User Stories/FR-NFR lists count as input sources. Page rendering: PDF = actual page count; DOCX = rendered count; Pasted text = 1 page per 3,000 chars (rounded up). Skipped image-only pages count toward limits. Total project cap: 100 pages.
Requirement Statement	An individually identified statement produced by splitting input text (e.g., R-001, R-002; unique in project, never reused). Original labels (e.g., "FR3") are preserved as metadata labels.
Extracted Item	An actor, entity/class, process, data flow, data store, interaction, or relationship derived from requirement statements. Tagged as system-extracted, user-added, or user-edited.
Shared Requirements Model	The project's unified requirement statements plus confirmed extracted items. Every diagram element references items from this model by ID.
Flag	A finding attached to a requirement statement: Ambiguous, Incomplete, Inconsistent, or Contradictory. Status: Open or Resolved. Flags are non-blocking.
Warning	A visible indicator on a diagram element derived from an open requirement flag or diagram notation rule validation.
Assumption	A diagram element that has no directly linked requirement statement (DR2). Visually distinguished across editor and exports.
Version	A saved state of a diagram created on generation, manual save, full regeneration, or revert. Keeps the latest 20 versions per diagram.
Active / Inactive User	Active: Authenticated request in last 1.5 min. Inactive: No login or activity for 30 consecutive days (triggers auto-purge workflows).

1.4 ROLES & PERMISSIONS MATRIX

System / Workspace Action	Viewer	Editor	Owner
Open project; view requirements, extracted items, diagrams, warnings; search; export diagrams	✓	✓	✓
Report issue or feedback (FR27a/b); view own reports	—	✓	✓
Upload/delete documents; add/edit requirement statements and extracted items; confirm extraction	—	✓	✓
Generate, regenerate, edit, save, customise, revert diagrams; resolve flags	—	✓	✓
Rename project; delete project; invite/assign/remove collaborators	—	—	✓
Access coverage summary, combined report, activity log, all feedback reports on project	—	—	✓

* Permissions are strictly enforced on the server (FR24e). System Administrators manage accounts globally and only access project content when explicitly added as a project member.

1.5 SYSTEM CONSTRAINTS

- **C1 (AI Provider):** Isolated behind module TE4 (Google Gemini Flash free tier). Quotas and rate limits apply. Multi-key failover is enabled per provider terms.
- **C2 (Hosting):** Frontend on Vercel; backend on Render free tier. External persistent storage is used for state; an uptime monitor calls health checks every 10 mins.
- **C3 (Storage Cap):** System-wide storage cap defaults to 1 GB.
- **C4 (Course Boundaries):** Custom encryption key management, formal penetration testing, and accessibility certifications are out of scope.

2 FUNCTIONAL REQUIREMENTS (FR1 – FR11)

2.1 Requirement & Document Ingestion

ID	Requirement Specification Details
FR1	Accept requirements via typed/pasted text, uploaded PDF or DOCX documents (max 10 MB per file), existing user stories, and FR/NFR lists. Input language is restricted to English.
FR2	Validate input before processing. Reject corrupted, oversized, password-protected files, or PDFs with no extractable text. For PDFs containing both text and image-only pages, extract text pages and issue explicit warnings for skipped unreadable pages. Never fail silently.
FR3	Support project management: create, name, rename, list, switch, and delete projects. Project names are unique per owner (case-insensitive). Limit combined project inputs to 15 documents or 100 total pages.

2.2 Requirement Analysis & Information Extraction

ID	Requirement Specification Details
FR4	Split inputs into uniquely identified requirement statements (ID and source line reference). Extract modeling elements: actors, entities/classes, processes, data flows, data stores, interactions, and relationships. Maintain direct back-links to source requirement IDs.
FR5	Provide viewing, editing, addition, deletion, and explicit confirmation of extracted information before generation. Subsequent requirement modifications clear confirmation. Generation is blocked without server-verified active confirmation.

2.3 Ambiguity & Requirement Validation

ID	Requirement Specification Details
FR6	Detect ambiguous, incomplete, inconsistent, and contradictory statements; flag affected requirements with categorized findings.
FR7	Highlight flagged requirements, explain root cause in plain language, present clarifying questions, and offer suggested fixes for manual or automated resolution.
FR8	Allow diagram generation with unresolved flags (non-blocking status). Display active warning badges directly on affected diagram elements.

2.4 Diagram Recommendation & Generation

ID	Requirement Specification Details
FR9	Analyze requirements to recommend diagram types with supporting rationale. Allow user selection overrides prior to final confirmation.
FR10	Generate Use Case, Class, Activity, Sequence, DFD, and Component diagrams from confirmed requirements with live progress tracking and timestamp logging.
FR11	Send in-app notifications and optional emails upon generation success or failure, including when users navigate away.

2. FUNCTIONAL REQUIREMENTS (FR1 2 – FR22)

2.5 Diagram Review & Interactive Editing

ID	Requirement Specification Details
FR1 2	Present generated diagrams as editable drafts supporting element additions, removals, renames, repositioning, and property tuning. Saving creates a new version.
FR1 3	Support selective partial regeneration or full diagram regeneration. Full regeneration creates an automated pre-regeneration backup version snapshot.
FR1 4	Propagate modified source requirements to affected diagrams upon approval. Retain a version history of the latest 20 versions per diagram with single-click rollback.
FR1 5	Provide visual styling adjustments (colour schemes, layout orientation, font sizing) without altering underlying structural model relationships.

2.6 Traceability, Consistency & Export

ID	Requirement Specification Details
FR1 6	Maintain a single shared requirements model behind all diagrams. References are linked dynamically so model updates propagate across traceability views.
FR1 7	Provide derivation explanations for diagram elements; flag unrepresented requirements; notify users of requirement drift; detect cross-diagram naming inconsistencies and prompt for unified synchronization.
FR1 8	Export diagrams in PNG, SVG, or A4 PDF formats with responsive label fitting, including a quick one-click default format export feature.

2.7 Guidance, Management & Reporting

ID	Requirement Specification Details
FR1 9	Guide users step-by-step through the pipeline with dismissible onboarding guides and contextual notation helpers.
FR20	Provide project-level coverage metrics: mapped requirements, unresolved flags, and unrepresented requirement counts.
FR21	Allow Owners to compile and export a unified PDF project report containing requirements, diagrams, and open issue logs.
FR22	Maintain an immutable project activity log tracking requirement edits, diagram saves, regenerations, and reversions.

2. FUNCTIONAL REQUIREMENTS (FR23 – FR30)

2.8 Administration, Collaboration & System Integration

ID	Requirement Specification Details
FR23	Enable System Administrators to view, add, deactivate, and reactivate user accounts. Account deactivation revokes access while preserving project data.
FR24	Support collaborator invitations with defined roles (Viewer, Editor, Owner). Server enforces role permissions per request. Conflict detection warns on stale state saves.
FR25	Provide an administrative monitoring dashboard showing active users, storage utilization, generation times against NFR1, and error logs (read-only, no secrets).
FR26	Provide multi-filter search engine supporting keyword, ID, status (flagged, unresolved, exported), and diagram type filters (AND across categories, OR within).
FR27	Allow users to submit contextual feedback or issue reports on specific requirements/diagrams, visible to Owners and System Administrators.
FR28	Support Email/Password registration (with verification) and Google OAuth. Implement password reset links (1 hr validity) and 15-minute account lockouts after 5 failed attempts.
FR29	Automatically validate generated diagrams against notation standards (DR1) and extracted items post-generation; highlight structural warnings and repair options.
FR30	Handle API rate limits via queueing and automated retry (up to 3 retries in 5 mins). Display user-friendly startup screens during cold backend boots without losing input data.

3. NON-FUNCTIONAL REQUIREMENTS (NFR1 – NFR5)

ID	Category	Non-Functional Metric & Performance Target
NFR1	Performance	For a typical 20-page document (~8,000 words), selected diagram generation shall achieve a median latency ≤ 60 s and 95th percentile ≤ 120 s. Exceedances are logged for admin review.
NFR2	Usability	First-time users shall complete upload, generation, and export workflows unaided in ≤ 1 hour (verified by $\geq 80\%$ pass rate in 5-user usability testing).
NFR3	Security & Encryption	All uploaded documents and project database records must be encrypted at rest using AES-256 and in transit via HTTPS/TLS.
NFR4	Access Control	Authenticate sessions with automated 24-hour inactivity timeout. Server-enforced role access control returns HTTP 404 for unauthorized project requests.
NFR5	Privacy & Data Retention	Provide in-app Privacy Policy outlining AI provider data transmission. Require user consent before processing. Automatically delete inactive account data after 30 days (warnings sent days 25–30). Support immediate data self-deletion. Account-less active projects transfer ownership to active Editors. Retain anonymized audit logs for 90 days.

3. NON-FUNCTIONAL REQUIREMENTS (NFR6 – NFR10)

ID	Category	Non-Functional Metric & Performance Target
NFR6	Maintainability & Efficiency	Support incremental diagram regeneration. Modifying a single requirement statement (affecting ≤ 10 diagram elements) shall complete within 30 seconds.
NFR7	Availability	Achieve ≥ 95% system uptime during weekday operating hours (Mon–Fri, 08:00–20:00 IST). Scheduled system maintenance is restricted to Saturdays and Sundays.
NFR8	Compatibility	Ensure full functional alignment on desktop Google Chrome, Mozilla Firefox, and Microsoft Edge (latest 2 stable major releases) at resolutions ≥ 1280px.
NFR9	Scalability	Sustain performance metrics for at least 5 concurrent diagram generation requests without violating NFR1 targets; additional requests queue sequentially.
NFR10	Workspace Security	Provide secure multi-user collaborative workspaces with granular role enforcement (FR24) across all endpoints.

4. DOMAIN REQUIREMENTS (DR1 – DR2)

ID	Domain Specification Details
DR1	Model software artifacts adhering strictly to recognized standard engineering notations: UML Use Case, Class, Activity, Sequence, and Component diagrams, and Data Flow Diagrams (distinguishing processes, data flows, data stores, and external entities with distinct standardized visual symbols).
DR2	Build diagram elements exclusively from extracted requirements. When software requirements lack explicit structural details for necessary diagram elements, mark those visual elements explicitly as Assumptions , styling them distinctly from verified requirement elements across canvas rendering and all export formats (PNG, SVG, PDF).